HOMEWORK FOR PROBABILITY AND STATISTICS

PART 1: PROBABILITY

Problem 1. Urn A contains 3 red and 3 black balls, whereas urn B contains 4 red and 6 black balls. A ball is randomly selected from each urn.

(a) What is the probability that the two balls have the same color?

(b) What is the probability that the two balls have different colors?

Problem 2. A certain system can experience three different types of defects. Let A_i $(i = 1, 2, 3)$ denote the event that the system has a defect of type i. Suppose that $P(A_1) = 0{,}12$; $P(A_2) = 0{,}07$; $P(A_3) = 0{,}05$; $P(A_1 \cup A_2) = 0{,}13$; $P(A_1 \cup A_3) = 0{,}14$; $P(A_2 \cup A_3) = 0{,}10$; $P(A_1 \cap A_2 \cap A_3) = 0{,}01$.

(a) What is the probability that the system does not have a type 1 defect?

(b) What is the probability that the system has both type 1 and type 2 defects?

(c) What is the probability that the system has both type 1 and type 2 defects but not a type 3 defect?

(d) What is the probability that the system has at most two of these defects?

Problem 3. Three coins are tossed simultaneously. The probability of obtaining a tail on the first coin is $\frac{1}{3}$; the probability of obtaining a tail on the second coin is $\frac{2}{5}$; and the third coin is fair.

(a) Compute the probability that at least one coin shows a tail.

(b) Compute the probability that exactly two coins show tails.

Problem 4. A box contains 5 blue balls, 6 red balls, and 2 yellow balls, all of which have the same mass and size. Three balls are drawn at random from the box. Determine the probability that:

(a) no red ball is selected;

(b) the selected balls contain more than one color.

Problem 5. A company manufactures telecommunication devices in three different workshops. The defect rates in these workshops are 1%, 1.5%, and 3%, respectively. To determine an appropriate warranty period for the electronic devices, a product is randomly selected from the storage warehouse and inspected.

(a) Compute the probability that the inspected device is non-defective, given that the proportions of products from the three workshops in the warehouse are $3 : 3 : 4$.

(b) After the inspection, it is found that the randomly selected device is defective. Determine the probability that this defective device originated from the second workshop.

Problem 6. A box contains 10 six-sided dice of identical shape and size, consisting of the following types: 6 fair and homogeneous dice (Type A); 2 dice for which the probability of rolling a 6 is $\frac{1}{12}$ (Type B); and 2 dice for which the probability of rolling a 6 is $\frac{1}{4}$ (Type C). A player randomly selects one die from the box and rolls a 6. Which type of die is the player most likely to have used?

Problem 7. Consider 3 urns. Urn A contains 2 white and 4 red balls, urn B contains 8 white and 4 red balls, and urn C contains 1 white and 3 red balls. If 1 ball is selected from each urn, what is the probability that the ball chosen from urn A was white given that exactly 2 white balls were selected?

Problem 8. A fair and homogeneous die is rolled twice consecutively. Compute the probability that the sum of the numbers of dots obtained in the two rolls is a prime number.

Problem 9. From a box containing 12 balls, including 6 red balls, 4 blue balls, and 2 yellow balls, three balls are drawn at random. Compute the probability that the three selected balls are of exactly two different colors.

Problem 10. Six balls are to be randomly chosen *at the same time* from an urn containing 8 red, 10 green, and 12 blue balls.

(a) What is the probability at least one red ball is chosen?

(b) Given that no red balls are chosen, what is the conditional probability that there are exactly 2 green balls among the 6 chosen?

Problem 11. Urn A contains 2 white and 4 red balls, whereas urn B contains 1 white and 1 red ball. A ball is randomly chosen from urn A and put into urn B, and a ball is then randomly selected from urn B. What is

(a) the probability that the ball selected from urn B is white?

(b) the conditional probability that the transferred ball was white given that a white ball is selected from urn B?

Problem 12. Suppose that 3 percent of men and 0,5 percent of women are color blind. A color-blind person is chosen at random. What is the probability of this person being male? Assume that there are an equal number of males and females. What if the population consisted of twice as many males as females?

Problem 13. On rainy days, Joe is late to work with probability 0,3; on non-rainy days, he is late with probability 0,1. With probability 0,7, it will rain tomorrow.

(a) Find the probability that Joe is early tomorrow?

(b) Given that Joe was early, what is the conditional probability that it rained?

Problem 14. Let X be a discrete random variable with the following cumulative distribution function

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{3} & \text{if } 0 \le x < 1 \\ \frac{2}{5} & \text{if } 1 \le x < 2 \\ \frac{3}{5} & \text{if } 2 \le x < 3 \\ \frac{9}{10} & \text{if } 3 \le x < 3{,}5 \\ 1 & \text{if } 3,5 \le x. \end{cases}$$

(a) Find the probability mass function of random variable X.

(b) Find $P\left(\frac{3}{2} < X \le 4\right)$.

(c) Find $E(X)$, $V(X)$.

Problem 15. Let X be a discrete random variable that has distribution table as follows

X	1	2	3
$P(X=k)$	p_1	p_2	p_3

Furthermore, $\begin{cases} p_1 + 2p_2 + 3p_3 = \frac{13}{6} \\ p_1 + 4p_2 + 9p_3 = \frac{31}{6} \end{cases}$.

(a) Find $E(X^2)$ and $V(X)$.

(b) Find $P(2 \leq X < 4)$.

Problem 16. Let X be a discrete random variable that has distribution table as follows

X	1	2	3
$P(X = k)$	p_1	p_2	p_3

Also assume that $\begin{cases} p_1 + 2p_2 + 3p_3 = \frac{13}{6} \\ p_1 + 4p_2 + 9p_3 = \frac{31}{6} \end{cases}$.

(a) Find $P(X \geq 2)$.

(b) Determine the cumulative distribution function of random variable X.

Problem 17. Two coins are to be flipped. The first coin will land on heads with probability 0,6; the second with probability 0,7. Assume that the results of the flips are independent, and let X equal the total number of heads that result.

(a) Find $P(X = 1)$?

(b) Find $E(X)$?

Problem 18. Customers at a gas station pay with a credit card (A), debit card (B), or cash (C). Assume that successive customers make independent choices, with $P(A) = 0{,}5$; $P(B) = 0{,}2$ and $P(C) = 0{,}3$.

(a) Among the next 100 customers, what are the mean and variance of the number who pay with a debit card?

(b) Answer part (a) for the number among the 100 who don't pay with cash?

Problem 19. Suppose that the number of accidents in one year on a certain highway is a Poisson random variable. If the average number of accidents yearly on this highway is 2,2, determine the probability such that

(a) no accident in the next year;

(b) exactly 3 accidents in the next year;

(c) at least 2 accidents in the next year;

Problem 20. Assume that X is a Poisson random variable with $\lambda = 1$.

(a) Find $P(X > 3)$?

(b) Find $P(X \geq 1 | X \leq 3)$?

Problem 21. Let X be a continuous random variable that has the following density function

$$f(x) = \begin{cases} kx & \text{if } 0 \leq x \leq 1 \\ k & \text{if } 1 \leq x \leq 4 \\ 0 & \text{if } x \notin [0, 4]. \end{cases}$$

(a) Find $E(X)$ and $V(X)$.

(b) Find $P(0 < X \leq 3)$?

Problem 22. Assume that the density function of the continuous random variable X as follows

$$f(x) = \begin{cases} ax + bx^2 & \text{if } x \in (0, 1), \\ 0 & \text{if } x \notin (0, 1), \end{cases}$$

and $E(X) = 0{,}6$.

(a) Find $P(X < 0{,}5)$?

(b) Find $V(X)$?

Problem 23. Let

$$f(x) = \begin{cases} kx^2(1 - x), & x \in [0, 1] \\ 0 & x \notin [0, 1] \end{cases}$$

be the density function of the continuous random variable X.

(a) Find $V(X)$.

(b) Find $P(X > 0, 4)$.

(c) Determine the median of X?

Problem 24. Assume that the density function of the continuous random variable X as follows

$$f(x) = \begin{cases} kx(x^2 - 1) & \text{if } x \in [1, 4] \\ 0 & \text{if } x \notin [1, 4]. \end{cases}$$

Construct the cumulative distribution function of X and determine $E(X)$.

Problem 25. The time (in hours) required to repair a machine is an exponentially distributed random variable with parameter $\lambda = 1/2$. What is

(a) the probability that a repair time exceeds 2 hours?

(b) the conditional probability that a repair takes at least 10 hours, given that its duration exceeds 9 hours?

Problem 26. The lifespan of a light bulb is a continuous random variable X with the following probability density function

$$f(x) = \begin{cases} \dfrac{a}{x^2} & \text{if } x \geq 400 \text{ (hours)} \\ 0 & \text{if } x < 400 \text{ (hours).} \end{cases}$$

(a) Determine the constant a?

(b) Find the proportion of products with a lifespan greater than 800 hours?

(c) Observe the lifespan of 100 light bulbs. Let Y be the number of light bulbs with a lifespan greater than 1000 hours. What is the distribution law of Y? Find $E(Y)$ and $V(Y)$?

Problem 27. If X is a normal random variable with parameters $\mu = 10$ and $\sigma^2 = 4$, compute

(a) $P(X > 12)$

(b) $P(9 < X < 12)$

(c) $P(X < 8)$.

Problem 28. Suppose that X is a normal random variable with mean 5. If $P(X > 9) = 0{,}2$, approximately what is σ^2?

Problem 29. Suppose that X is a normal variable with mean 12 and variance 4.

(a) Find $P(10 < X < 13)$?

(b) Find the value of c such that $P(X > c) = 0{,}1$?

Problem 30. If 65 percent of the population of a large community is in favor of a proposed rise in school taxes, approximate the probability that a random sample of 100 people will contain

(a) at least 50 who are in favor of the proposition;

(b) between 60 and 70 inclusive who are in favor;

(c) fewer than 75 in favor.

Problem 31. Suppose that 3 balls are randomly selected from an urn containing 3 red, 4 white, and 5 blue balls. If we let X and Y denote, respectively, the number of red and white balls chosen, then what is the joint probability mass function of X and Y?

Problem 32. Let X, Y be jointly random variables with the joint probability mass function $f(x, y) = c(x + y)$, where $x = 1, 2, 3, 4$ and $y = 1, 2, 3$. Determine the distribution table and find $P(|X - Y| > 1)$.

Problem 33. Let X be a discrete random variable that takes on the values $0, 1, 2$ with equal probability and Y be a random variable with a binomial distribution with parameters $n = 3$ and $p = 0{,}8$. Assume that X and Y are independent.

(a) Determine the joint distribution table of X and Y.

(b) Find $P(X - Y > 0)$.

PART 2: STATISTICS

For convenience, we recommend that students use calculators in this section.

Problem 34. Blood pressure values are often reported to the nearest 5 mmHg (100, 105, 110, ...). Suppose the actual blood pressure values for nine randomly selected individuals are

118,6; 127,4; 138,4; 130,0; 113,7; 122,0; 108,3; 131,5; 133,2.

What is the sample mean and sample standard deviation of the reported blood pressure values?

Problem 35. In a large university, the following are the ages of 20 randomly chosen employees:

24	31	28	43	28	56	48	39	52	32
38	49	51	49	62	33	41	58	63	56

Assuming that the data come from a normal population, construct a 95% confidence interval for the population mean μ of the ages of the employees of this university.

Problem 36. A drug is suspected of causing an elevated heart rate in a certain group of high-risk patients. Twenty patients from the group were given the drug. The changes in heart rates were found to be as follows.

-1	8	5	10	2	12	7	9	1	3
4	6	4	12	11	2	-1	10	2	8

Construct a 98% confidence interval for the mean change in heart rate. Assume that the population has a normal distribution

Problem 37. A random sample of size 36 is drawn from a population having a normal distribution. The sample mean and the sample standard deviation from the data are given, respectively, as $\bar{x} = -2{,}22$ and $s = 1{,}67$. Construct a 98% confidence interval for the population mean μ.

Problem 38. Let a random sample of size 100 from a normal population for which both mean μ and variance σ^2 are unknown yield $\bar{x} = 3{,}12$ and $s^2 = 1{,}04$. Determine a 99% upper confidence bound for μ.

Problem 39. An auto manufacturer gives a bumper-to-bumper warranty for 3 years or 36000 miles for its new vehicles. In a random sample of 120 of its vehicles, 40 of them needed five or more major warranty repairs within the warranty period. Estimate the true proportion of vehicles from this manufacturer that need five or more major repairs during the warranty period, with confidence coefficient 95%.

Problem 40. In a random sample of 500 items from a large lot of manufactured items, there were 40 defectives. Find a 90% confidence interval for the true proportion of defectives in the lot.

Problem 41. For a health screening in a large company, the diastolic and systolic blood pressures of all the employees were recorded. In a random sample of 150 employees, 12 were found to suffer from hypertension. Find 98% lower bound for the proportion of the employees of this company with hypertension.

Problem 42. A random sample of size 20 is drawn from a population having a normal distribution. The sample mean and the sample standard deviation from the data are given, respectively, as $\bar{x} = 5{,}2$ and $s = 1{,}4$. Construct a 90% confidence interval for the population variance σ^2.

Problem 43. A random sample of 25 observations gave the following summary statistics: $\sum x_i = 234$ and $\sum x_i^2 = 3048$. Assuming that the data can be looked upon as a random sample from a normal population, construct a 95% confidence interval for the actual variance σ^2.

Problem 44. At a brewery, the amount of beer automatically filled into bottles follows a normal distribution with a mean of 500ml. If the variance of the amount of beer filled into bottles is high, many bottles will either be underfilled or overflow. An engineer measures the amount of beer in 20 randomly selected bottles, obtaining a sample standard deviation of $s = 5$. Construct a 95% confidence interval for the variance of the beer filling process.

Problem 45. Observe the lifetimes of a number of light bulbs with the results given in the following table. Assume that the lifetimes of the light bulbs follow a normal distribution. Construct a 98% confidence interval for the average lifetime of this type of light bulb. If a 95% confidence interval with a length not exceeding 30 hours is required, what is the minimum sample size needed?

Lifetime (h)	1000	1050	1100	1130	1150	1170	1200	1240
Number of bulbs	3	5	6	5	4	8	6	5

Problem 46. The melting point of each of 25 samples of a certain brand of hydrogenated vegetable oil was determined, resulting in $\bar{x} = 94{,}32$. Assume that the distribution of the melting point is normal with $\sigma = 1{,}2$. Test $H_0 : \mu = 95$ versus $H_a : \mu \neq 95$ using a two tailed level 0,01 test.

Problem 47. A random sample of 50 measurements resulted in a sample mean of 62 with a sample standard deviation 8. It is claimed that the true population mean is at least 64. Is there sufficient evidence to refute the claim at the 2% level of significance?

Problem 48. Consider the test $H_0 : \mu = 35$ and $H_a : \mu > 35$ for a population that is normally distributed. A random sample of 18 observations taken from this population produced a sample mean of 40 and a sample standard deviation of 5. Using $\alpha = 0{,}05$, would you reject the null hypothesis?

Problem 49. A manufacturer claims that the mean life of batteries manufactured by his company is at least 44 months. A random sample of 40 of these batteries was tested, resulting in a sample mean life of 41 months with a sample standard deviation of 16 months. Test at $\alpha = 0{,}01$ whether the manufacturer's claim is correct.

Problem 50. A random sample is given from a normal distribution as follows.

69	65	72	73	59	55	39	52	67	57
56	50	70	47	56	45	70	64	67	53

For $\alpha = 0{,}05$, could we conclude that $\mu = 55$?

Problem 51. A machine in a certain factory must be repaired if it produces more than 12% defectives among the large lot of items it produces in a week. A random sample of 175 items from a week's production contains 25 defectives, and it is decided that the machine must be repaired. Does the sample evidence support this decision? Use $\alpha = 0{,}02$.

Problem 52. It is claimed that two of three Americans say that the chances of world peace are seriously threatened by the nuclear capabilities of other countries. If in a random sample of 400 Americans, it is found that only 252 hold this view, do you think the claim is correct? Use $\alpha = 0{,}05$.

Problem 53. Consider the hypothesis test $H_0 : \mu_1 = \mu_2$ against $\mu_1 \neq \mu_2$ with known variances $\sigma_1 = 10$ and $\sigma_2 = 7$. Suppose that sample sizes $n_1 = 22$ and $n_2 = 25$ and that $\bar{x}_1 = 5$, $\bar{x}_2 = 8$. Test the hypothesis. Use $\alpha = 0{,}05$.

Problem 54. The following information was obtained from two independent samples selected from two normally distributed populations with unknown but equal variances. Test at the $\alpha = 0{,}025$ significance level whether μ_1 is lower than μ_2.

Sample 1	14	15	11	14	10	8	13	10	12	16	15	17	12
Sample 2	17	16	21	12	20	18	16	14	21	20	13	20	13

Problem 55. Consider the hypothesis test $H_0 : \mu_1 = \mu_2$ against $\mu_1 < \mu_2$ with unknown variances but $\sigma_1^2 = \sigma_2^2$. Assume that $s_1^2 = 4$ and $s_2^2 = 6{,}25$. Futhermore, suppose that sample sizes $n_1 = 15$ and $n_2 = 15$ and that $\bar{x}_1 = 6{,}5$, $\bar{x}_2 = 8$. Test the hypothesis, use $\alpha = 5\%$.

Problem 56. Consider two sample from normal distributions with unknown variances but $\sigma_1^2 = \sigma_2^2$. Assume that $s_1^2 = 4$ and $s_2^2 = 6{,}25$. Futhermore, suppose that sample sizes $n_1 = n_2 = 36$ and that $\bar{x}_1 = 4$, $\bar{x}_2 = 8$. Test the hypothesis, use $\alpha = 5\%$. Test at the $\alpha = 0{,}05$ significance level whether μ_1 is lower than μ_2.

Problem 57. Two different types of injection-molding machines are used to form plastic parts. A part is considered defective if it has excessive shrinkage or is discolored. Two random samples, each of size 300, are selected, and 15 defective parts are found in the sample from machine 1 while 8 defective parts are found in the sample from machine 2. Is it reasonable to conclude that both machines produce the same fraction of defective parts, use $\alpha = 0{,}05$.

Problem 58. A random sample of 500 adult residents of Maricopa County found that 385 were in favor of increasing the highway speed limit to 75 mph, while another sample of 400 adult residents of Pima County found that 267 were in favor of the increased speed limit. Do these data indicate that there is a difference in the support for increasing the speed limit between the residents of the two counties. Use $\alpha = 0{,}05$.

Problem 59. The following are midterm and final examination test scores for 10 students from a calculus class, where x denotes the midterm score and y denotes the final score for each student. Find the least-squares regression line for these data.

x	68	87	75	91	82	77	86	82	75	79
y	74	79	80	93	88	79	97	95	89	92

Problem 60. The following data give the annual incomes (in thousands of dollars) and amounts (in thousands of dollars) of life insurance policies for eight persons.

Annual income	42	58	27	36	70	24	53	37
Life insurance	150	175	25	75	250	50	250	100

(a) Estimate the sample correlation coefficient? What can we conclude about the relationship between the annual incomes and amounts?

(b) Find the simple linear regression model? Suppose that someone has and annual income of 65 (in thousand of dollars), using this linear model to estimate the amount of life insurance policies for this person?